Statistical Code Review Report – ClassIQ

1. Introduction

This report presents the statistical code review of the ClassIQ project using SonarQube and SonarScanner. The aim was to evaluate code quality, identify issues, and analyze metrics such as coverage, complexity, and duplication.

2. Tools Used

SonarQube – Static analysis platform
SonarScanner – Analysis execution tool
Maven – Build and test tool
JaCoCo – Code coverage measurement

3. Summary Metrics

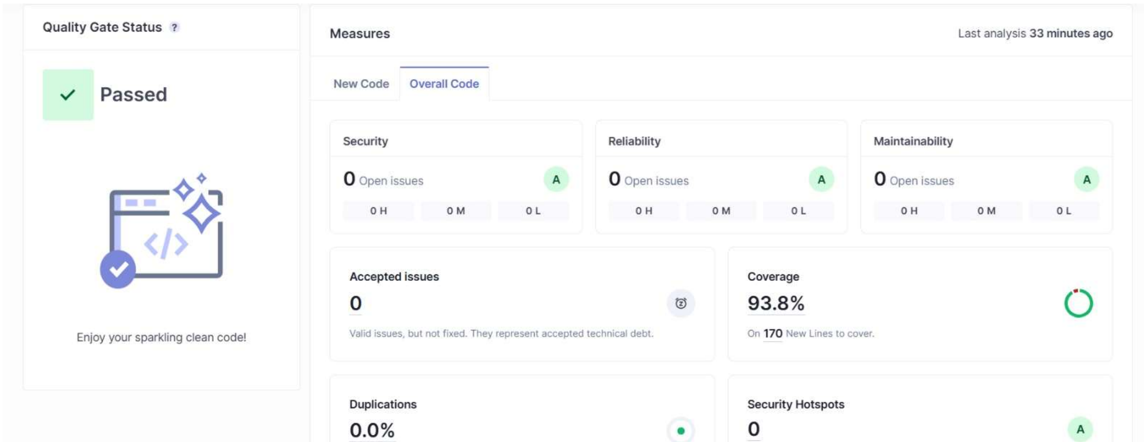
Metric	Value
Quality Gate	Passed
Coverage	93.8%
Security Issues	0
Reliability Issues	0
Maintainability	A
Duplications	0%

4. Interpretation of Results

The results indicate that the project meets high-quality standards. The coverage of 93.8% demonstrates strong test effectiveness. The absence of security, reliability, and maintainability issues confirms that the code is stable and well-structured.

5. Visual Evidence

Figure 1: SonarQube Dashboard showing Quality Gate Passed and Coverage



6. High-Priority Issues Identified

- Generic exception usage
- Duplicate literals in code
- Low initial test coverage
- Resource handling issues

All these issues were fixed during the development process.

7. Recommendations

Issue	Recommendation
Exception Handling	Use custom exceptions
Code Duplication	Use constants
Test Coverage	Add more unit tests
Complexity	Refactor large methods

8. Conclusion

The final code review confirms that the ClassIQ project meets all required quality standards. The project achieved a passing Quality Gate, high coverage, and no critical issues.